

# **Software Development Life Cycle (SDLC)**

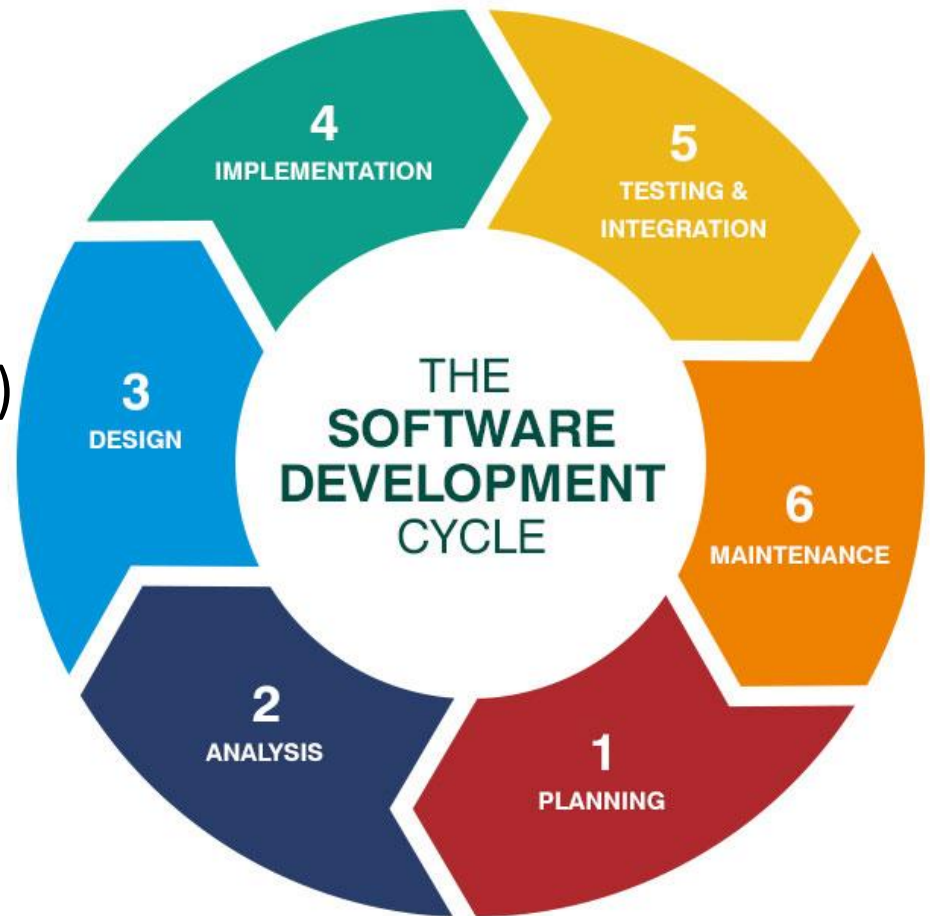
- A systematic process used to develop high-quality software.
- It defines phases involved in planning, creating, testing, and maintaining software.

## **What is SDLC?**

- SDLC is a framework that describes all phases involved in software development.
- It ensures structured and organized development.

# ***SDLC Phase's***

1. Requirement Analysis
2. System Design
3. Implementation (Coding)
4. Testing
5. Deployment
6. Maintenance



# **Phase 1: Requirement Analysis**

- Collect requirements from users or clients.
- Understand what the system should do.
- Document all requirements clearly.

# **Phase 2: System Design**

- Design system architecture.
- Prepare database design.
- Plan UI and system structure.

## **Phase 3: Implementation (Coding)**

- Developers write actual code.
- Use programming languages to build software.
- Convert design into working system.

## **Phase 4: Testing**

- Test software for bugs and errors.
- Perform different testing types.
- Ensure software works correctly.

## **Phase 5: Deployment**

- Release software to users.
- Install system in real environment.
- Make it available for actual use.

## **Phase 6: Maintenance**

- Fix bugs after deployment.
- Improve features.
- Update system when needed.

- SDLC helps in structured development.
- Improves software quality.
- Ensures proper planning, development, testing, and maintenance.

*Thank You*